



Air University
(Mid Semester Examination: Spring 2025)

Student ID: _____

Subject: **Linear Algebra**
Course Code: **MA-201**
Class: **BS-CYS**
Semester: **II**
Section: **A, B (Evening Session)**

Total Marks: **50 Marks**
Date: _____
Time: _____
Duration: **2 Hours**
FM Name: **Mr. Amir Shahzad**

HoD's Signature: _____

FM's Signature: _____

Note:

- All questions must be attempted. Understanding the question is part of the examination.
- This examination carries 25% weight toward the final grade.
- Scientific calculator is allowed.
- Return the question paper with the answer sheet.
- Rough work or writing on the question paper will be considered as use of unfair means.

Q. No. 1 (CLO-1) (PLO-1)		20 Marks
a	Identify (determine) the value(s) of "λ" for which the system has. a) No solution. b) Exactly one solution. c) Infinitely many solutions. $\begin{aligned} x - 3y + 5z &= 1 \\ x + 3y - 3z &= 5 \\ 2x - y - (\lambda^2 - 4)z &= \lambda \end{aligned}$	7
b	Use the "Inversion Algorithm" to find the inverse of the given matrix A (if it exists). $E_1 = \begin{bmatrix} -1 & 2 & 4 \\ 0 & 6 & 3 \\ 1 & 5 & 6 \end{bmatrix}$	7
c	Solve the given system using the LU factorization for the matrix A $\begin{aligned} x + y + 2z &= 9 \\ 2x + 4y - 3z &= 1 \\ 3x + 6y - 5z &= 0 \end{aligned}$	6
Q. No. 2 (CLO-2) (PLO-1)		20 Marks
a	Boron sulphide reacts violently with water to form boric acid and hydrogen sulphide gas. The unbalanced equation is $B_2S_3 + H_2O \rightarrow H_3BO_3 + H_2S$	7
b	Find all "x" in R^4 that are mapped into the zero vector by the transformation $x \mapsto Ax$ for the given matrix A. $A = \begin{bmatrix} 1 & -4 & 7 & -5 \\ 0 & 1 & -4 & 3 \\ 2 & -6 & 6 & -4 \end{bmatrix}$	7
c	Assume that "T" is a linear transformation. Find the standard matrix of "T" such that $T: R^2 \rightarrow R^2$ rotates point (about the origin) through $\frac{3\pi}{2}$ radian (counterclockwise)	6

Q. No. 3 (CLO-4) (PLO-2)

10 Marks

Write equations that describe the flow, and then find the general solution of the system. Label the street intersections (junctions) and the unknown flows in the branches, as shown in Table 1. At each intersection, set the flow in equal to the flow out.

INTERSECTION	FLOW-IN	FLOW-OUT
A	$300+500$	$x_1 + x_2$
B	$x_4 + x_2$	$300 + x_3$
C	$100+400$	$x_4 + x_5$
D	$x_1 + x_5$	600

10

***** End *****